

EXPERIENCE

- Senior CVML Researcher / Engineer, Google XR** Apr'22 – Present
Computer Vision and Multimodal modeling - AI Interface team
 - Technical Lead:** Built an **LLM-aided Expert Helper** for guiding users to perform **procedural tasks** via **step-by-step** instructions
 - Developed a fusion approach via **prompt engineering** and **finetuning Gemini** for multi-turn multimodal conversations to improve visual understanding, state tracking, mistake detection, clarification accuracy & mistake correction. Patents filed.
 - Designed a **synthetic data generation** approach via prompting & **video generation** to scale dataset; finetuned models on it
 - Integrated and deployed the models end-to-end in **AR Glasses** through backend Python and **Kotlin** production servers.
 - Technical Lead:** Developed **unsupervised learning** algorithms for solving **language-vision tasks** using **Generative AI** models
 - Developed **zero-shot segmentation**, **semantic assignment**, **image caption retrieval** and **classification** techniques using the self and cross attention layers of **Stable Diffusion** backbone. **Open-sourced** the work on segmentation and semantics.
 - Our segmentation methods **outperformed SOTA** mean IOU for **unsupervised segmentation** by 17% and pixel accuracy by 26% over COCO-Stuff 27 benchmark. Published at **CVPR 2024**. Multiple patents filed.
 - Developed **semantic segmentation** models for **indoor scenarios** (data scarce) using Deep-Lab V3+ architecture.
 - Augmented synthetic datasets with Segment Anything (SAM) masks to synthesize high quality training data and improved perf.
 - Reduced training data size by 95% (2M to 110k samples), while improving segmentation mean IOU from 64.95% to 65.55%.
 - Developed and deployed **phone as a controller** capability for **AR devices** via training **phone-hand gesture detection** models
 - Trained Tap-Net architecture NN models for real-time gesture (phone turns, taps, double taps) identification on large datasets (>200k samples) establishing > 99% detection rate, <0.2% gesture confusion and <0.8% false positive rate. Patent filed.
- Applied Scientist II, Amazon** Dec'20 – Apr'22
Applied Scientist I, Amazon Mar'19 – Nov'20
Computer Vision Research - Amazon One (AO)
 - Developed computer vision models for AO, a recognition device utilizing **palm biometrics**, for identification and payments.
 - Built the primary **image quality estimation** modules along with developing **anomaly detection** models by fusing classical CV approaches with low latency neural network training. Worked cross-functionally to iterate & finalize next-gen hardware specs.
 - Reduced end to end modeling pipeline latency by 7% while maintaining accuracy parity via multi-task network learning.
 - Shipped models in live system for **2020 AO launch**. AO is available at 500+ Whole Foods stores & 150+ 3rd party locations today.
- Applied Scientist Intern, Amazon Alexa ML** Summer 2018
Language Modeling and Speech Recognition - Amazon Alexa
 - Developed a **Domain Adaptation** technique for **LSTM Language Models** to improve generalization of unseen Alexa domains
 - Trained a domain independent LM using adversarial loss function; augmented the model with domain embeddings via attention
 - Improved LM perplexity by 20% for a 0.7 times smaller model, embeddings obtained via TF-IDF scores, autoencoders and LDA
- Research Intern, EPFL Switzerland** Summer 2016
Social Network Models for Brain Networks, Advisors - Prof. Nisheeth Vishnoi and Dr. Elisa Celis
 - Developed a graphical model based on game theory, for the neuron-connections in animal brains, using constrained max-flows
 - Evaluated the model and obtained a fit better than present models; Scaled it 50x for larger networks via Hadoop-Map Reduce
- Research Intern, IIT Kanpur, India** Summer 2015
Localization of Audio Sources using Signal Processing - Multimodal Information Processing Systems Lab
 - Developed an algorithm inspired by human auditory mechanism, to determine the **position of a sound source in 3D** space
 - Enhanced the directional information using an additional sensor with head related transfer functions for broadband sources
 - Publication accepted and presented at IEEE European Signal Processing conference (**EUSIPCO 2016**)

PUBLICATIONS and PATENTS

- Diffuse, Attend and Segment: Unsupervised Zero-Shot Segmentation using Stable Diffusion:** Junjiao Tian, Lavisha Aggarwal, Andrea Colaco, Zsolt Kira, MGFranco. Proceedings of IEEE Conference on Computer Vision and Pattern Recognition (**CVPR 2024**)
- Identity preserving loss for learned image compression:** Jiahong Xiao, Lavisha Aggarwal, Prithviraj Banerjee, Manoj Aggarwal, Gerard Medioni. Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPRW 2022**)
- Binaural source localization using a HRTF data model with enhanced frequency diversity:** CS Reddy, Rishika Agarwal, Lavisha Aggarwal, Rajesh M Hegde. 2016 24th European Signal Processing Conference (**EUSIPCO 2016**)
- YETI (YET to Intervene) Proactive Interventions by Multimodal AI Agents in Augmented Reality Tasks:** Saptarashmi Bandyopadhyay, Vikas Bahirwani, Lavisha Aggarwal, Bhanu Guda, Lin Li, Andrea Colaco (Arxiv 2025)
- Fairness in AI Systems: Mitigating gender bias from language-vision models:** Lavisha Aggarwal, Shruti Bhargava (Arxiv 2018)
- Lavisha Aggarwal, Manoj Aggarwal, Gerard Guy Medioni, Dilip Kumar. System for generating enhanced training data:** U.S. Patent No. 12,190,566. (**U.S. Patent and Trademark Office**). January 2025

- Junjiao Tian, [Lavisha Aggarwal](#), Mar Gonzalez, Andrea Colaco. **Unsupervised zero-shot segmentation mask generation and semantic labeling**: U.S. Patent No. 18/792,206. (**U.S. Patent and Trademark Office**). February 2025

EDUCATION

University of Illinois at Urbana-Champaign	Master of Science - Computer Science	2017 – 2019	GPA - 4.0/4.0
Indian Institute of Technology Kanpur	Bachelor of Tech - Electrical Engineering, Minor - Computer Science - Artificial Intelligence	2013 - 2017	GPA - 9.4/10.0

AWARDS AND FELLOWSHIPS

- Secured **All India Rank 324** among 1.4 million students in **JEE Mains 2013** (AIEEE equivalent)
- Secured **All India Rank 594** among 150 thousand students in **IIT JEE Advanced 2013**
- **KVPY Fellowship 2012** by DST, Government of India for excellence in Science and Mathematics in Class 12th
- **Academic Excellence Award - 2013, 2014, 2016** from **Indian Institute of Technology Kanpur** for outstanding academic performance
- **Ericsson Innovation Award 2015** from **Indian Institute of Technology Delhi** for top 5 projects nationwide and awarded INR 25000
- Invited for various summer research Internship programs - **SN Bose Scholars 2016** (UW Madison, IUSSTF), **Viterbi India 2016** (University of Southern California), **Summer at EPFL 2016** (Ecole Polytechnic Federale de Lausanne, Switzerland)

MENTORING

- JiuHong Xiao: Applied Science Intern at Amazon One, 2021, pursuing PhD at NYU Tandon School of Engg.
- Junjiao Tian: Summer Research Intern at Google XR, 2023, working as Research Scientist at Meta Generative AI
- Saptarashmi Bandyopadhyay: Student Researcher at Google XR, 2024, pursuing PhD at University of Maryland
- Qin Liu: Student Researcher at Google XR, 2025, joining as Postdoctoral Scholar at Stanford University
- Akshay Paruchuri: Student Researcher at Google XR, pursuing PhD at University of North Carolina, Chapel Hill

TEACHING EXPERIENCE

- **Graduate Teaching Assistant – Computer Vision CS 543, UIUC** with Prof Svetlana Lazebnik **Spring 2018**
- **Graduate Teaching Assistant – Applied Machine Learning CS 498, UIUC** with Prof David Forsyth **Fall 2018**
- **Academic Mentor – Electrodynamics PHY 103, IIT Kanpur**: Took EE remedial lectures and individually tutored **2** students **Fall 2014**

TECHNICAL SKILLS

Languages: Python, Kotlin, C++, C, MATLAB, SQL, Go, Javascript, R **Libraries:** TensorFlow, PyTorch, Keras, MXNet, OpenCV

ACADEMIC SERVICE

- **Steering Committee Member** - Multi-Agent AI in the Real World Workshop, **AAAI 2025**
- **Reviewer** - ICCV, CVPR Workshops on CV for Mixed Reality & Synthetic Data for CV, IEEE Trans. on Info. Forensics and Security
- **DUGC Student Representative** - Electrical Engg. **IIT Kanpur, 2016** - Mediated EE student grievances with department faculty

GRAD SCHOOL PROJECTS

- **Clustering High Dimensional Data** **Feb'18 – Dec'18**
 - Formulated a clustering algorithm for unknown number of clusters, by enriching the data with isotropy, local spatial properties
 - Learnt property weights via backpropagation and developed a Gibbs energy minimization framework for obtaining clusters
- **Image Retrieval - Deep Learning and BOW (Vocabulary Tree)** **Nov'17 – Dec'17**
 - Clustered SIFT features hierarchically and ranked similar images by TF-IDF scores in a bag of visual words model
 - Trained VGG16 CNN network to learn a feature representation for retrieving the most similar image using cosine, neural loss
- **Computer Vision - Online Facial Clustering in Videos** **Jul'17 – Dec'17**
 - Developed an online-learning system to identify different people in a movie clip to count number of unique characters
 - Modeled facial features using GMM's and developed novel GMM update and comparison techniques
- **3D Display and User Interface** **Dec'14 – May'15**
 - Developed a C++ desktop application to display objects in 3D in real time by tracking head position and using perspective geometry
 - Awards - Adjudged among the top 5 projects at [Ericsson Innovation Awards, India – 2015](#) (A prototype for ecommerce-websites)
- **Removing bias from Image Captioning Models** **Jan'18 – Apr'18**
 - Established the presence of Gender bias in current models and developed a multi-task setup using a gender prediction network
 - Achieved a less biased image captioning system, while simultaneously providing captions more relevant to the image
- **Image Processing - Object Removal in a Photo via Image Inpainting** **Oct'16 – Nov'16**
 - Improved over Criminisi's inpainting approach through an adaptive regularizer and histogram based patch selection criterion
 - Achieved more "realistic" looking images. Adjudged as the Best Class Project and accepted at **SIGNAL - 2017**

ADVANCED COURSEWORK

Computer Vision, Computational Photography, Image Processing, ML in Signal Processing, Digital Signal Processing, Artificial Intelligence, Scientific Visualization, Optimization Techniques, ML in Natural Language Processing, Machine Learning, Distributed Systems